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COURSE PROJECT ASSIGNMENT

1. Project Overview

- **Team Composition: ~ 05 students**.
- **Topic:** Select a **real-world IT development project** (e.g., E-commerce platform, Mobile App, ERP module, or Health Information System).
- **Objective:** Apply the **PMBOK® Guide 6th Edition** framework to initiate, plan, execute, monitor, and close an IT project while delivering a functional software product.

2. Deliverable 1: The Lifecycle-Based Management Report

The report must follow the **Faculty Thesis Format**. The content must be structured chronologically according to the **5 Project Management Process Groups**, while integrating the **10 Knowledge Areas** using the **ITTO** (Inputs, Tools & Techniques, and Outputs) methodology.

Part I: Initiating Process Group

- **Project Initiation:** Develop the **Project Charter** to formally authorize the project.
- **Stakeholder Identification:** Identify all internal and external stakeholders and create a **Stakeholder Register**.

Part II: Planning Process Group This is the core of the report, requiring detailed plans for all knowledge areas:

- **Scope & Requirements:** Collect requirements, define the scope statement, and create a deliverable-oriented **Work Breakdown Structure (WBS)** and WBS Dictionary.
- **Schedule & Cost:** Define activities, sequence them (Network Diagram), perform **Critical Path Analysis**, and establish the **Cost Baseline**.
- **Quality & Resources:** Define quality metrics, testing plans, and the organizational structure using a **RACI Matrix**.
- **Risk & Communications:** Identify risks in a **Risk Register**, perform qualitative/quantitative analysis, and establish a communication management plan.

- **Procurement & Stakeholder Engagement:** Conduct make-or-buy analysis and define engagement strategies.

Part III: Executing Process Group

- **Project Work Management:** Describe how project work was directed and how project knowledge was managed.
- **Team Development:** Detail how the team was managed and developed (e.g., using the Tuckman model).
- **Quality & Communications:** Perform quality assurance and execute communication activities.

Part IV: Monitoring & Controlling Process Group

- **Integrated Change Control:** Demonstrate the process for managing changes to scope, schedule, or cost.
- **Performance Tracking:** Use **Earned Value Management (EVM)** to monitor project health and prevent scope creep.
- **Risk & Stakeholder Monitoring:** Detail how risks and stakeholder engagement were tracked and adjusted.

Part V: Closing Process Group

- **Project Closure:** Formalize product acceptance and transition.
- **Final Documentation:** Submit the final project report and an organized archive of all management artifacts.

*Note: This report must include a **"Lessons Learned"** summary highlighting the pros, cons, and practical reflections on applying PMBOK theory to your specific project.*

3. Deliverable 2: Live Presentation & Demos

- **Direct Presentation:** A **10-minutes** presentation using professional slides focusing on project value, management methodology, and results. A **5-minutes** Q&A.
- **Project Management Tool Demo:** Demonstrate the team's use of software (e.g., **Microsoft Project** for scheduling/tracking, **JIRA/Trello** for task management, or **GitHub** for version control).

4. Software Development Methodology

In addition to following the standard project management process groups, every team must select and implement a specific software development methodology. This choice must be clearly documented in the final report and demonstrated during the software presentation.

4.1. Methodology Selection & Justification

- **Requirement:** Teams must explicitly declare their choice of either the **Waterfall (Predictive)** or **SCRUM (Agile/Adaptive)** model during the planning phase.
- **Justification:** The report must justify this choice based on project characteristics such as requirement stability, team experience, risk levels, and stakeholder expectations. For instance, Waterfall is suitable for well-defined requirements, while SCRUM is preferred for evolving or complex scopes.

4.2. Model-Specific Requirements

A. Waterfall (Predictive) Model Requirements:

- **Sequential Execution:** The project must demonstrate linear progression through phases: Systems Analysis, Design, Construction, Testing, and Support.
- **Scope Baseline:** A detailed Work Breakdown Structure (WBS) is mandatory, organized either by project phases or product deliverables.
- **Formal Change Control:** A strict Change Log and an Integrated Change Control process must be established to prevent unauthorized scope creep.

B. SCRUM (Agile/Adaptive) Framework Requirements:

- **Defined Roles:** The team must assign specific roles: Product Owner (business value), ScrumMaster (process facilitator), and the Development Team (cross-functional execution).
- **Agile Artifacts:** Instead of a traditional WBS, teams must provide a **Product Backlog** prioritized by business value, a **Sprint Backlog** for the current cycle, and a **Burndown Chart** to visualize progress.
- **Ceremonies:** The report must include summaries or minutes from Sprint Planning, Daily Scrums, Sprint Reviews, and Sprint Retrospectives.

- **Iterative Delivery:** Teams must demonstrate the creation of "potentially shippable product increments" at the end of each designated Sprint.

4.3. Integrated Testing and Quality Assurance

Regardless of the chosen methodology, the project must include a systematic testing process:

- **Testing Levels:** Describe the execution of Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT).
- **Quality Metrics:** Define specific metrics to measure success, such as defect density, bug counts, or user satisfaction scores.

4.4. Tool Application

- **Waterfall Teams:** Encouraged to use **Microsoft Project** for Critical Path Analysis and maintaining a long-term Gantt chart.
- **SCRUM Teams:** Encouraged to use tools like **JIRA, Trello, or GitHub Projects** to manage the backlog and maintain a digital Kanban board.

4.5. Critical Reflection

In the "Lessons Learned" section of the final report, teams must compare their actual execution against the theoretical requirements of their chosen SDLC. They should specifically address challenges encountered, such as the difficulty of freezing requirements in Waterfall or the coordination effort required for Daily Scrums in an Agile environment.

- **Software Product Demo:** A live demonstration of the functional IT product. The software must solve a real-world business problem and include core management, search, and reporting features.

5. Work Division & Accountability

- **RACI Matrix:** Teams must provide a detailed Responsibility Assignment Matrix.
- **Contribution Percentage:** Quantify the actual contribution percentage (0-100%) for each of the 5 members to ensure fair and transparent grading.

6. Special Appendix: Managing the "Assignment Project"

Students are encouraged to view **the execution of this course assignment itself as a real project**. A dedicated Appendix must be included to demonstrate how the group managed their own work during the semester:

- **Scope, Schedule of the Assignment:** Defining all tasks required to complete the course requirements (not just the software).
- **Risk & Issue Management:** Detail how the group handled **actual difficulties** encountered during the term (e.g., member illnesses, technical environment errors, or time conflicts).
- **Change Management:** Describe how the group adapted their plan when internal or external changes occurred.
- **Reflective Lessons Learned:** Analyze what PMBOK theories worked well for the group's internal management and what challenges remain for future projects.